Storm Response Takes Flight

New helicopters and smarter drones are helping us respond faster and safer after major storms.

When severe weather strikes, aviation crews assess damage and deliver critical data and equipment to teams on the ground.

When storms cause outages, the top question from customers is: When will the power be back on?

Helping answer that question quickly – and safely – are the helicopters and drones of Duke Energy’s Utility Aviation and Unmanned Aerial Systems (UAS) teams. These “eyes in the sky” help our communities recover by enabling faster restoration after storms.

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After Hurricane Helene, helicopters transported personnel and supplies to flooded areas. Here, a utility pole is delivered to a ground team on Big Hungry Road in Flat Rock, N.C.

Each has a distinct role in storm response:

- Helicopters cover a lot of ground quickly to provide an initial assessment of the damage. They also have the endurance and lifting power to transport personnel and equipment to remote or flooded areas where roads are impassable.
- Drones offer speed and agility. Often, they're some of the first on-site with our lineworkers and damage assessment crews, flying into hazardous areas and capturing images to map damage and help plan repairs.

Together, they form a coordinated aerial strategy that improves operational efficiency through data collection, while keeping Duke Energy team members out of harm's way.

Helicopters: Strength in reach and lifting power

After storms, helicopters can cover hundreds of miles in a single day – far faster than ground crews. Their high-definition cameras and trained crews help locate downed lines, damaged crossarms, broken or leaning utility poles and washed-out access roads. This real-time data is critical as ground crews plan their coordinated response.

"These aircraft help us locate the problem, report it and respond faster," said Joey Santella, who oversees helicopter operations. "We'll fly on weekends and travel long distances to help get the power back on."

Helicopters also have the lifting power to transport personnel and heavy equipment to hard-to-reach areas. After Hurricane Helene in 2024, helicopters delivered utility poles, generators, fuel and other supplies to western North Carolina communities like Chimney Rock, where roads were washed out by flooding and mudslides.

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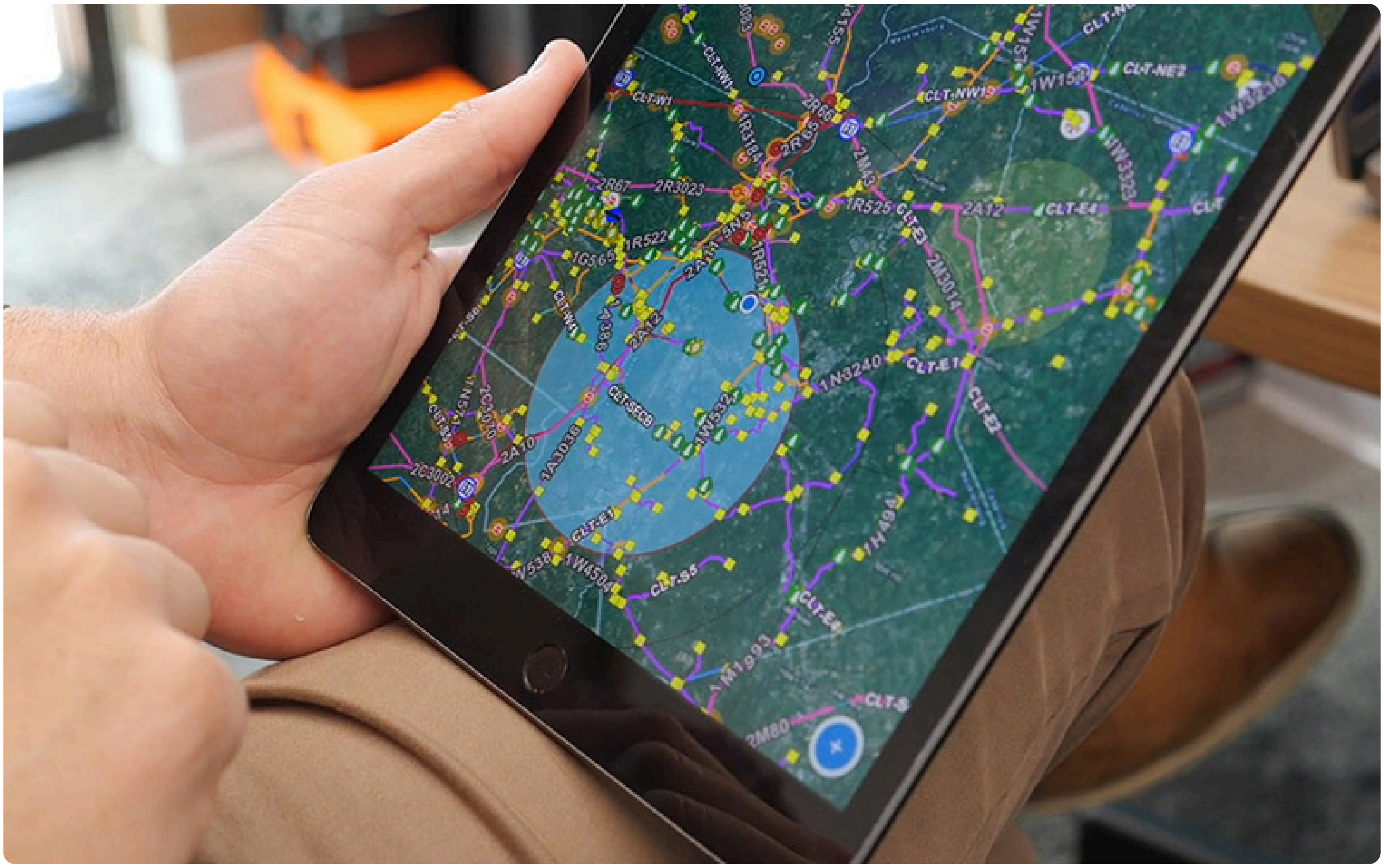

Using the AIR (Aviation Inspection and Routing) app, helicopter crews mark damage with GPS coordinates and assign priority levels to streamline communication with ground teams.

“You can’t drive something down a road that doesn’t exist,” said Luke Harper, who, with 16 years of storm experience, is director of Utility Aviation. “Our team logged hundreds of hours last year as we responded to hurricanes Debby, Helene and Milton.”

To enhance its response, Duke Energy developed an application named Aviation Inspection and Routing (AIR). The AIR app allows helicopter crews to mark damage with coordinates and assign priority levels to speed up repairs. GPS-tagged damage reports help ground crews navigate to the most critical areas and enhance safety by enabling pilots to flag aerial hazards.

What’s more: Duke Energy recently upgraded its helicopter fleet with a new Airbus H145 D3 – the first of four twin-engine aircraft. “These helicopters have greater lift capability and improved safety features to help increase the safety margin of our operation,” Harper said. “If you lose an engine, you’ve got a second one to keep you in the air.”

Drones: Precision, speed and safety

Duke Energy began using drones for storm response in 2015.

With experienced pilots like Isaac Medford of Duke Energy's Unmanned Aerial Systems (UAS) team, pictured, drones fly into hazardous or inaccessible areas after storms to capture detailed imagery for planning repairs.

What started with just two pilots has grown to more than 50, and the company now holds Federal Aviation Administration (FAA) authorization to fly beyond the visual line of sight – a major advancement that enables faster, remote response during large-scale events.

After Hurricane Helene, pilots used drones to help reconductor a transmission line in flooded Shelby, N.C. – stringing new wire across a river in just a few hours.

"That's impactful to customers," said Chief Drone Pilot Garrett Scott. "Transmission lines carry the bulk of power across the grid. They must come on line before we can restore electricity to homes and businesses."

In other cases, drones with infrared cameras will fly over solar fields to identify faulty panels. Pilots can determine if a panel is not converting sunlight efficiently by analyzing the color of their heat signatures. Then, technicians can make a repair.

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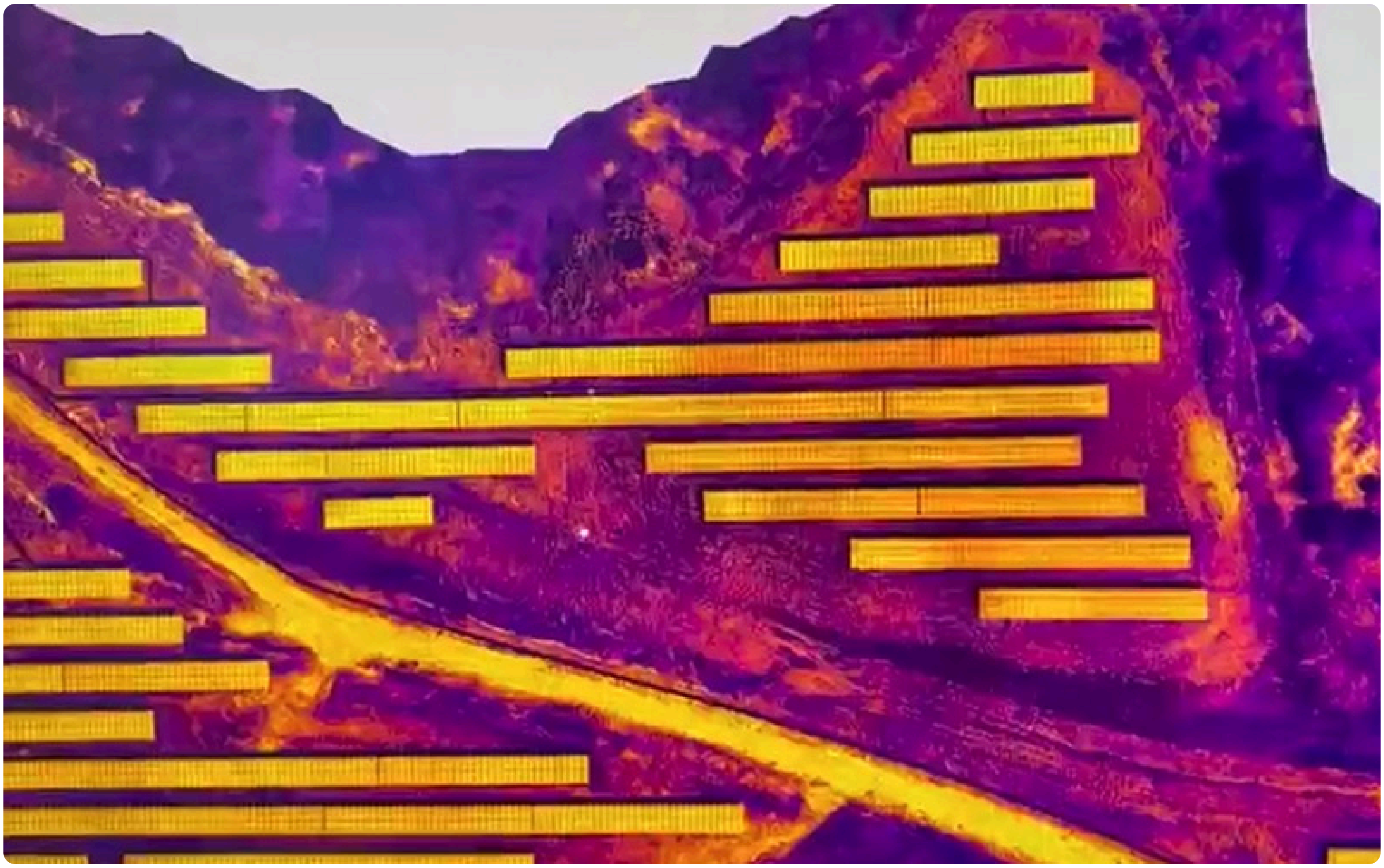

Using infrared cameras, drone pilots can identify damaged or underperforming solar panels, saving technicians the time-consuming task of walking the site to analyze panels with a handheld device.

Before drones, technicians would use a handheld thermal imaging device to test each section of a solar field, which can span hundreds of acres. So, this saves hours of post-storm inspection work.

Drones are also are used for environmental monitoring and confined space inspections, such as inside substations or nuclear power plants. In some cases, drones even collect water samples from hard-to-reach areas – keeping crews out of hazardous conditions.

"At its core, a drone is a flying camera, but drones can also sensor packages that capture different types of data in real time," said UAS Director Jackson Rollins. "Infrared cameras find heat signatures not visible to the naked eye and drones with LiDAR create 3D models of our sites. So, it's a versatile tool that helps teams across the company work more efficiently."

Veterans of the sky

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The Utility Aviation team at Duke Energy is a highly specialized group that plays a critical role in both routine grid maintenance and emergency storm response.

Many aviation team members bring emergency response or military experience.

Harper, who directs Utility Aviation, flew Blackhawk helicopters in Iraq and Afghanistan. Alex Justice, who coordinates drone crews, is a retired state trooper who used drones for crash-scene mapping.

Their shared mission now: helping Duke Energy customers recover faster from the unexpected.

"Whether it's a hurricane, ice storm or flood," Rollins said, "we're ready to respond – safely, quickly and efficiently."

Smarter tools for enhanced reliability

Based in Charlotte, N.C., Duke Energy's aerial teams work year-round to keep the grid strong. Their pre-storm patrols help Vegetation Management and Power Grid Operations prioritize work that benefits customers in the Midwest, Carolinas and Florida.

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In areas that aren't accessible with traditional methods, helicopter crews can trim trees from the sky using a 33-foot aerial saw.

For example, helicopters patrol more than 32,000 miles of transmission lines twice a year, looking for hazards like damaged equipment and trees or branches that are too close to power lines. This enables a proactive response before storms strike.

"We strive to improve grid reliability through both scheduled routine patrols and storm patrols," said Brian Anderson, chief pilot of Utility Aviation. "We are always on the lookout for new efficiencies and advancements to better serve our customers."

Drones complement helicopters with agility and precision. For example, drone crews examine poles and lines up close to look for loose connections or utility poles with structural damage. Drones are also used to validate the location of equipment and poles through GPS data, which informs future grid investments as Duke Energy builds a smarter energy future.

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